

CO2xNxPhrag: Porewater SO4/Cl

May 2026 Samples

2026-07-02

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-07-01" #Date that instrument was run
Run_by = "Zoe Read & Victoria Ellis" #Instrument user
Script_run_by = "Zoe Read & Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "May"
Site = "GCRew"
Project_name = "CO2xNxPhrag"
run_notes = "Stds 1 and 2 were low; need to remake them. Std curve looked amazing though.
" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/CO2xNxPhrag_202605_C1.txt"
raw_file_name_so4 = "Raw Data/CO2xNxPhrag_202605_S04.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xNxPhrag_PW_Processed_C1_S04_202605.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = "Raw Data/COMPASS_Synoptic_C1_S04_QAQClog_2026.csv"
```

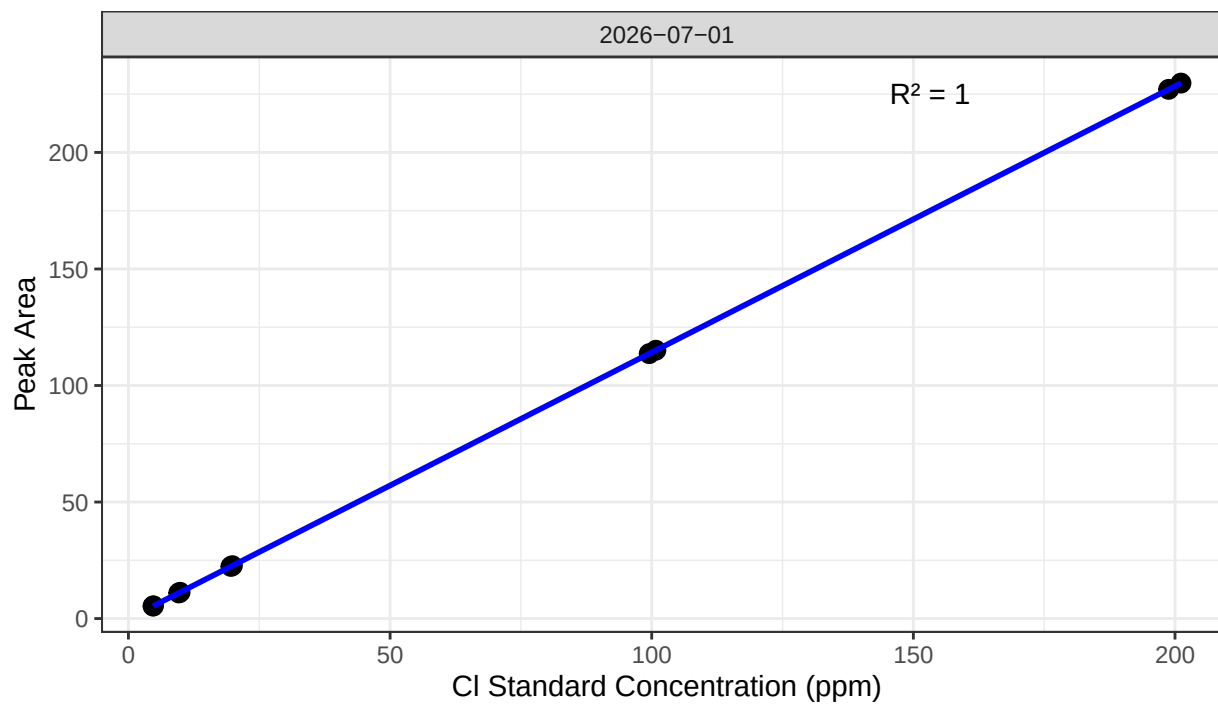
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

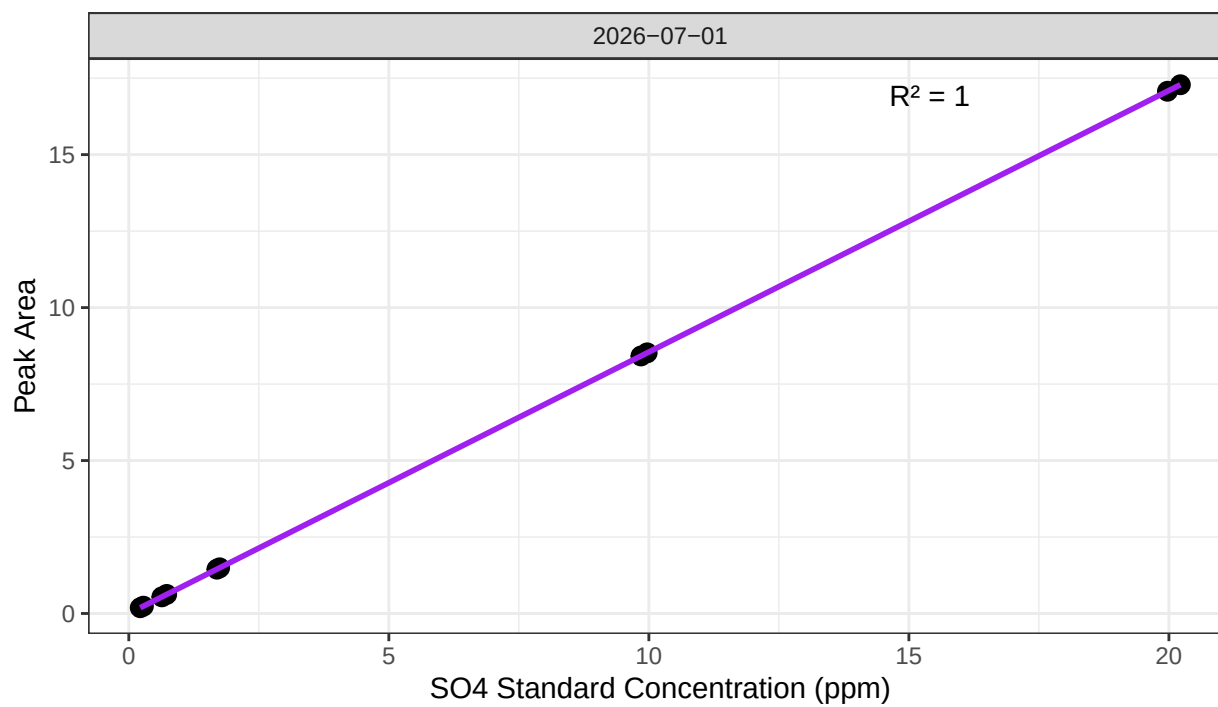
##Import Sample Data

0.2 Assess Standard Curves

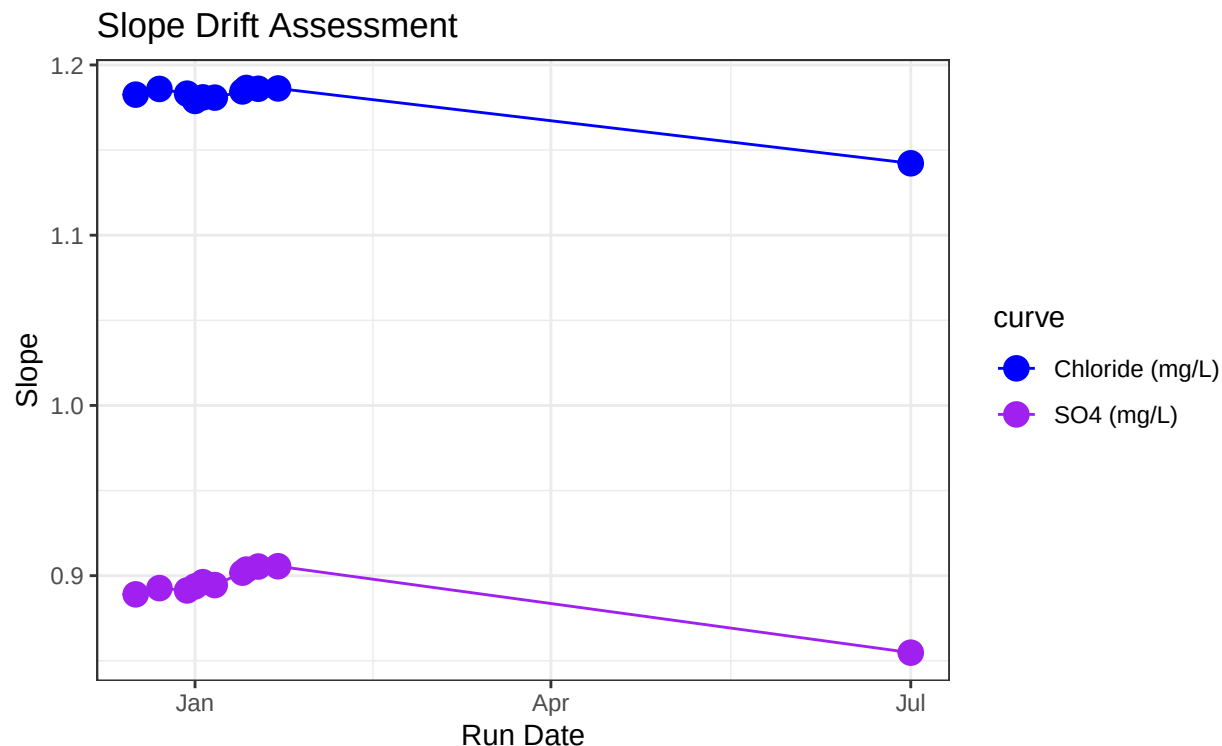
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

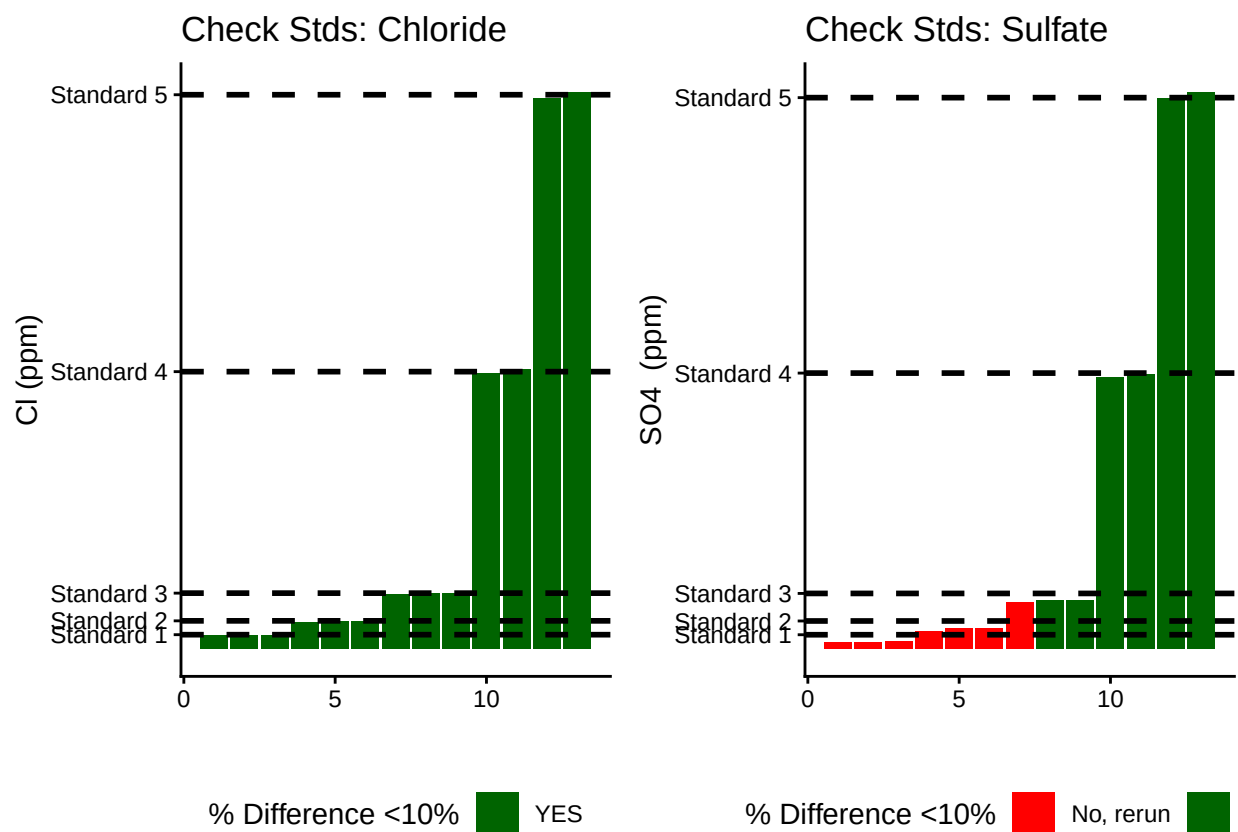
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 4.74 0.0437 0.00921 Chloride Check Standard RSD within Range - ~
## 2 Standard 2 9.77 0.130 0.0133 Chloride Check Standard RSD within Range - ~
## 3 Standard 3 19.8 0.166 0.00840 Chloride Check Standard RSD within Range - ~
## 4 Standard 4 100. 0.901 0.00899 Chloride Check Standard RSD within Range - ~
## 5 Standard 5 200. 1.66 0.00832 Chloride Check Standard RSD within Range - ~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.250 0.0302 0.121 Sulfate Check Standard RSD within Range - ~
## 2 Standard 2 0.700 0.0552 0.0789 Sulfate Check Standard RSD within Range - ~
## 3 Standard 3 1.73 0.0345 0.0199 Sulfate Check Standard RSD within Range - ~
## 4 Standard 4 9.91 0.0842 0.00850 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5 20.1 0.176 0.00875 Sulfate Check Standard RSD within Range - ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



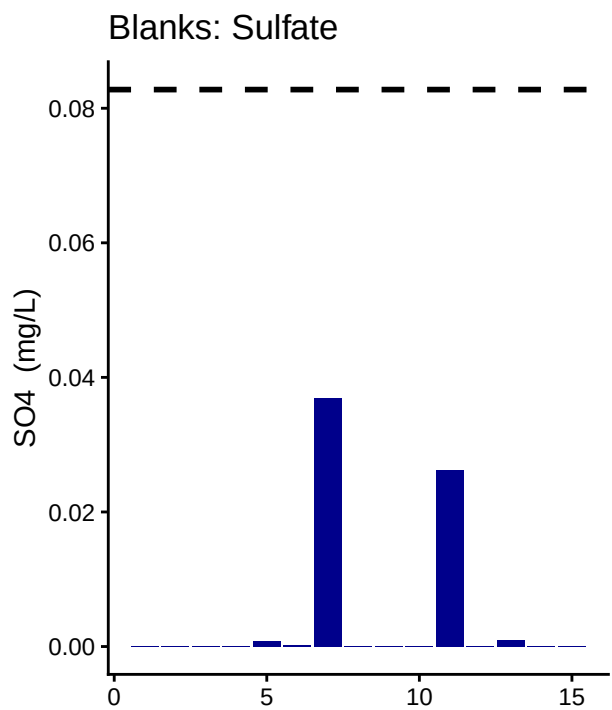
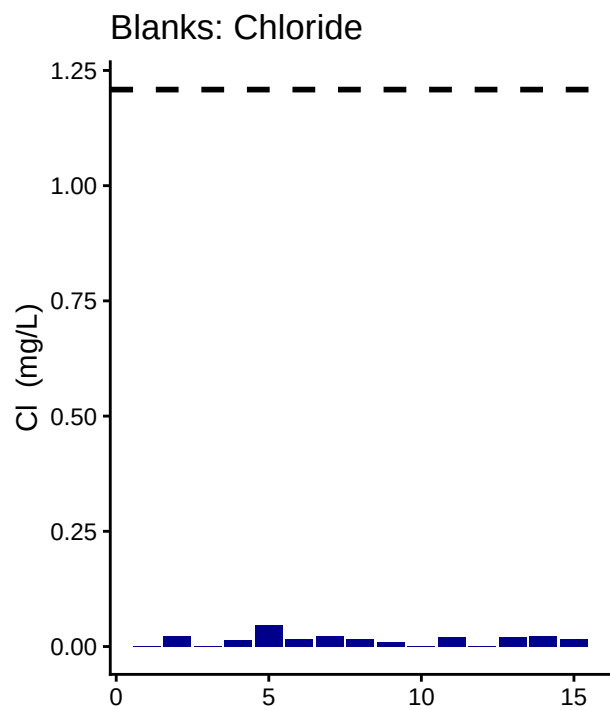
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] "<80% of Sulfate Check Standards are within range of expected concentration - REASSESS"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples █ Y

Blank Conc <25% Quartile Samples █ Y

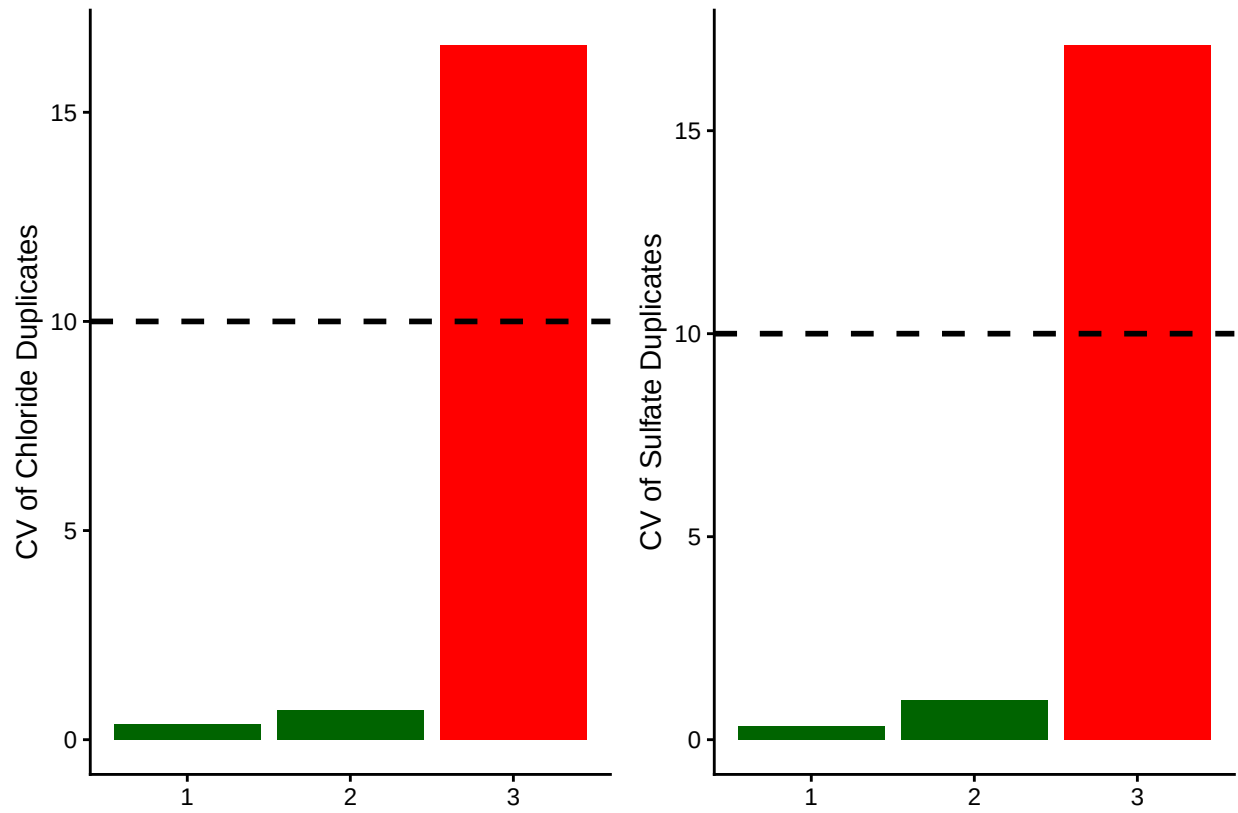
```
## Chloride blanks mean ppm:
```

```
## [1] 0.01504667
```

```
## Sulfate blanks mean ppm:
```

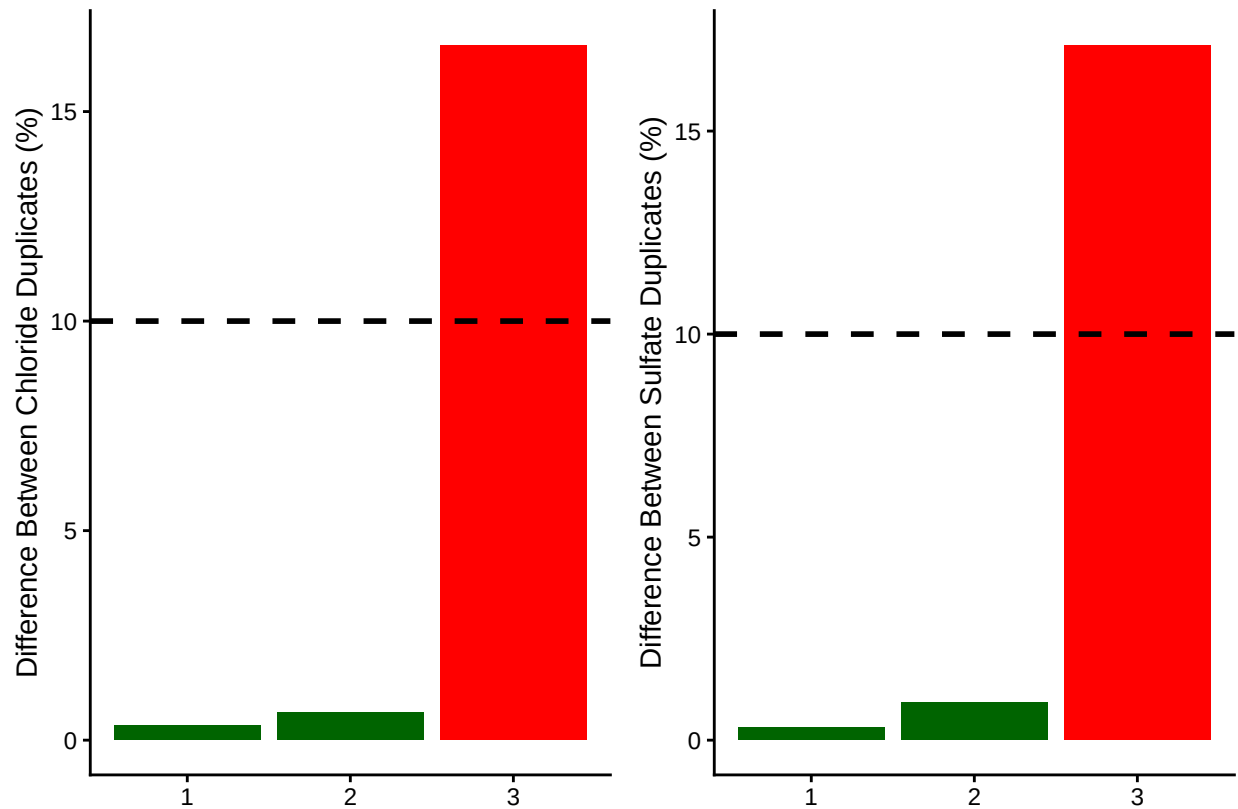
```
## [1] 0.004346667
```

0.5 Assess Duplicates



```
## [1] "<80% of Chloride Duplicates have a CV <10% - REASSESS"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```



```
## [1] "<80% of Chloride Duplicates have a CV <10% - REASSESS"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

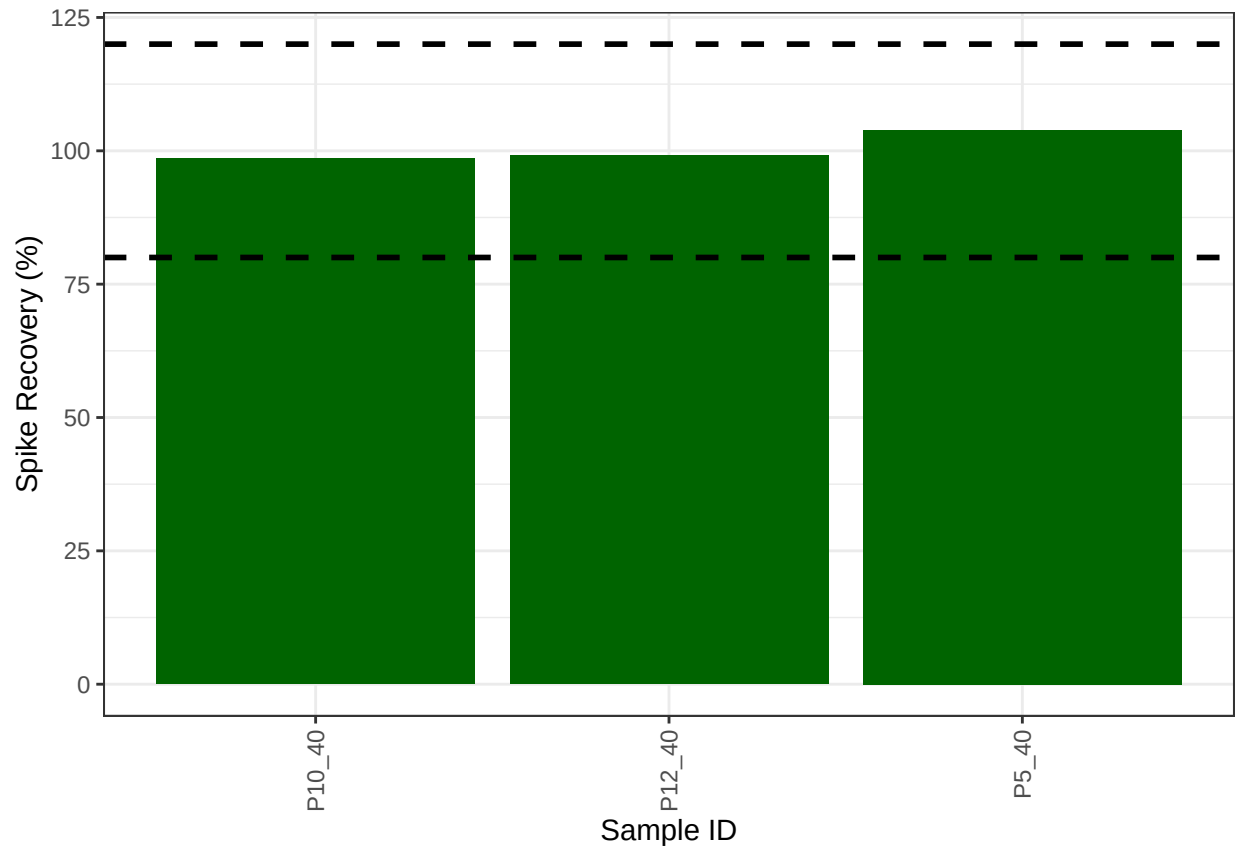
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```


0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

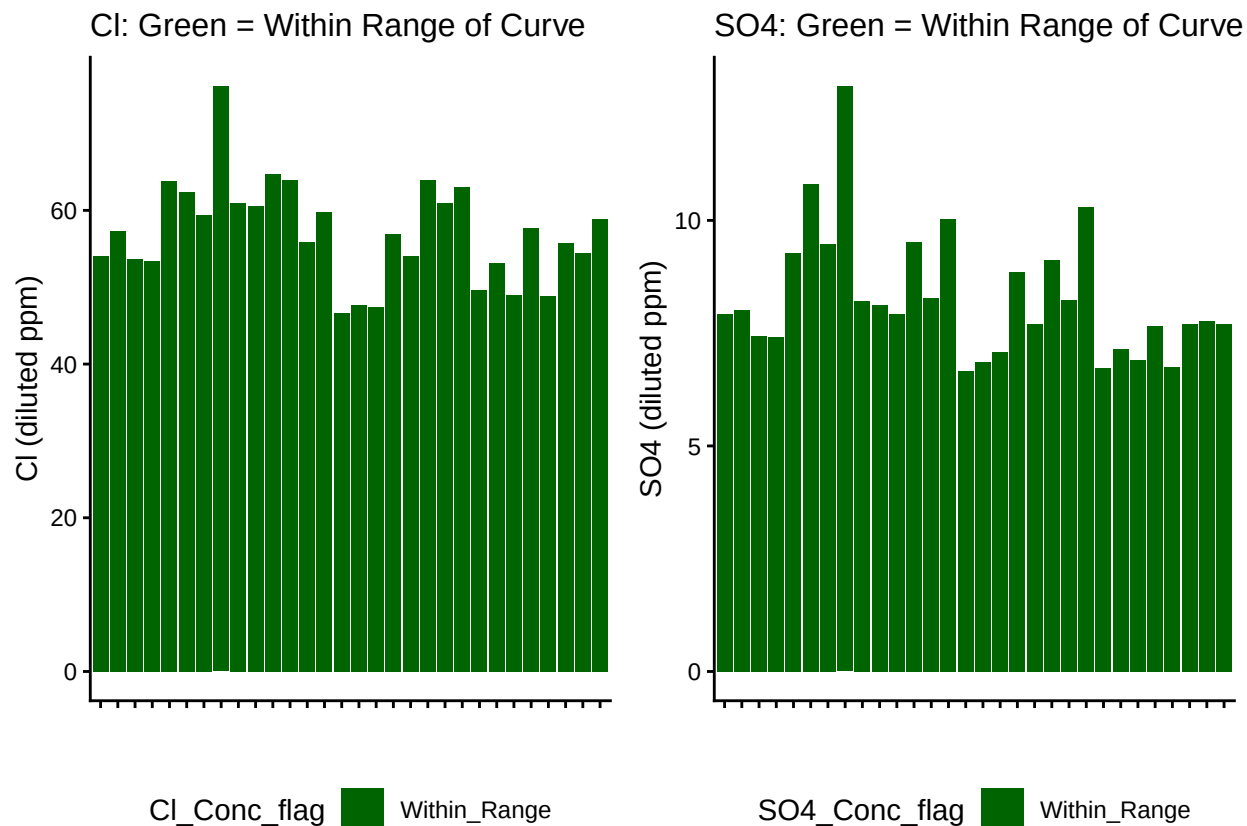


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

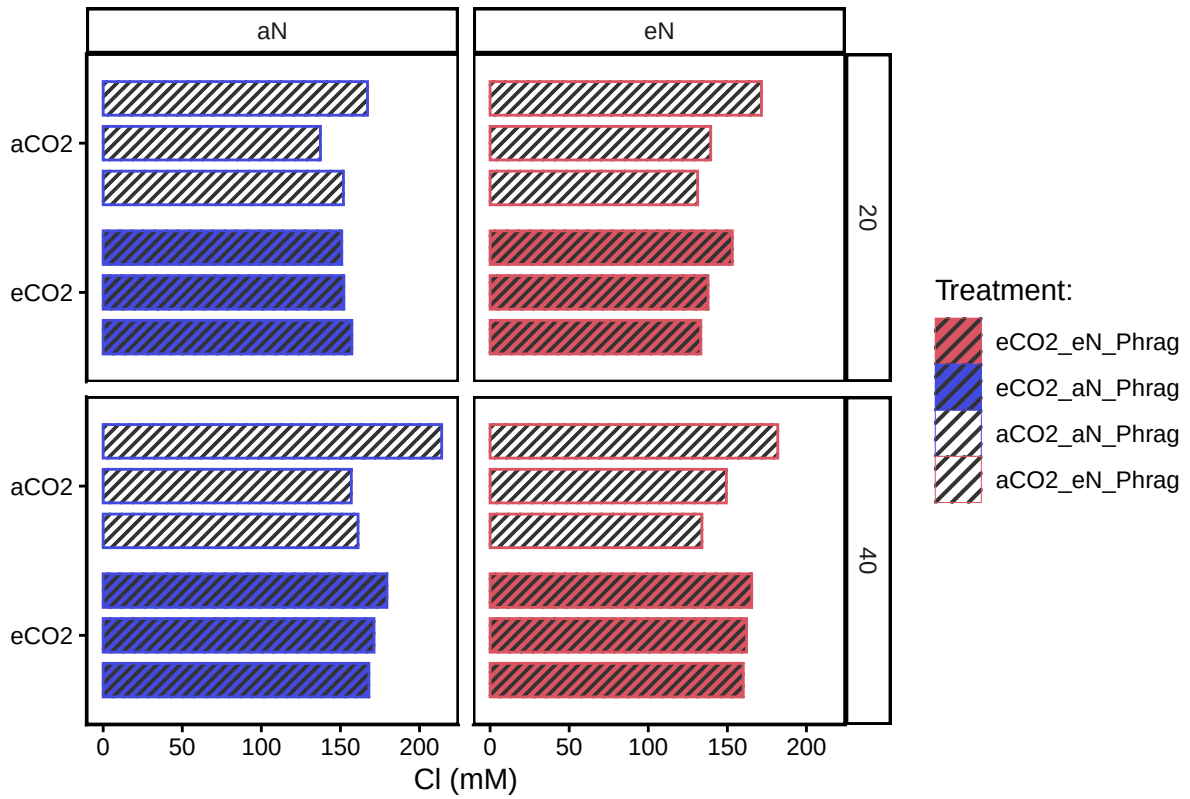
0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

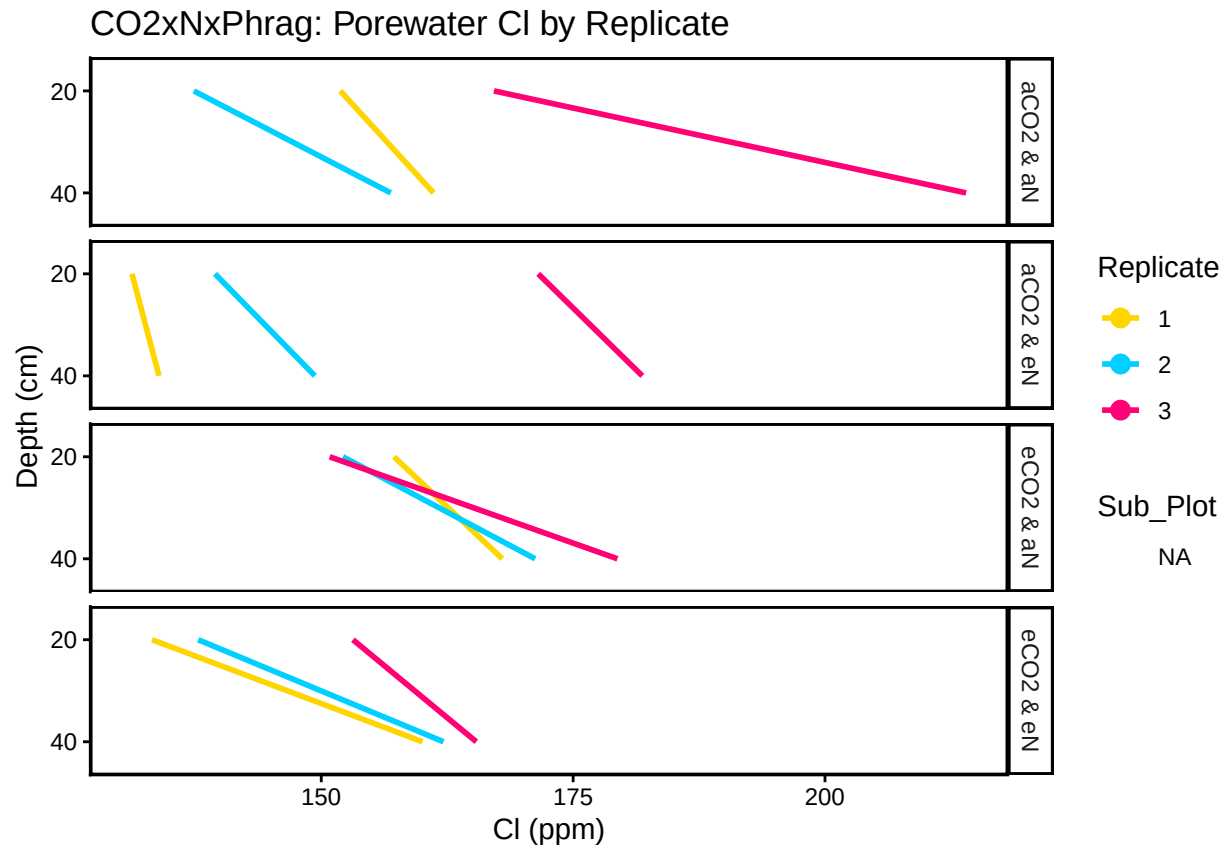
0.10 Visualize Data

```
## Visualize Data
```

CO2xNxPhrag: Porewater Chloride

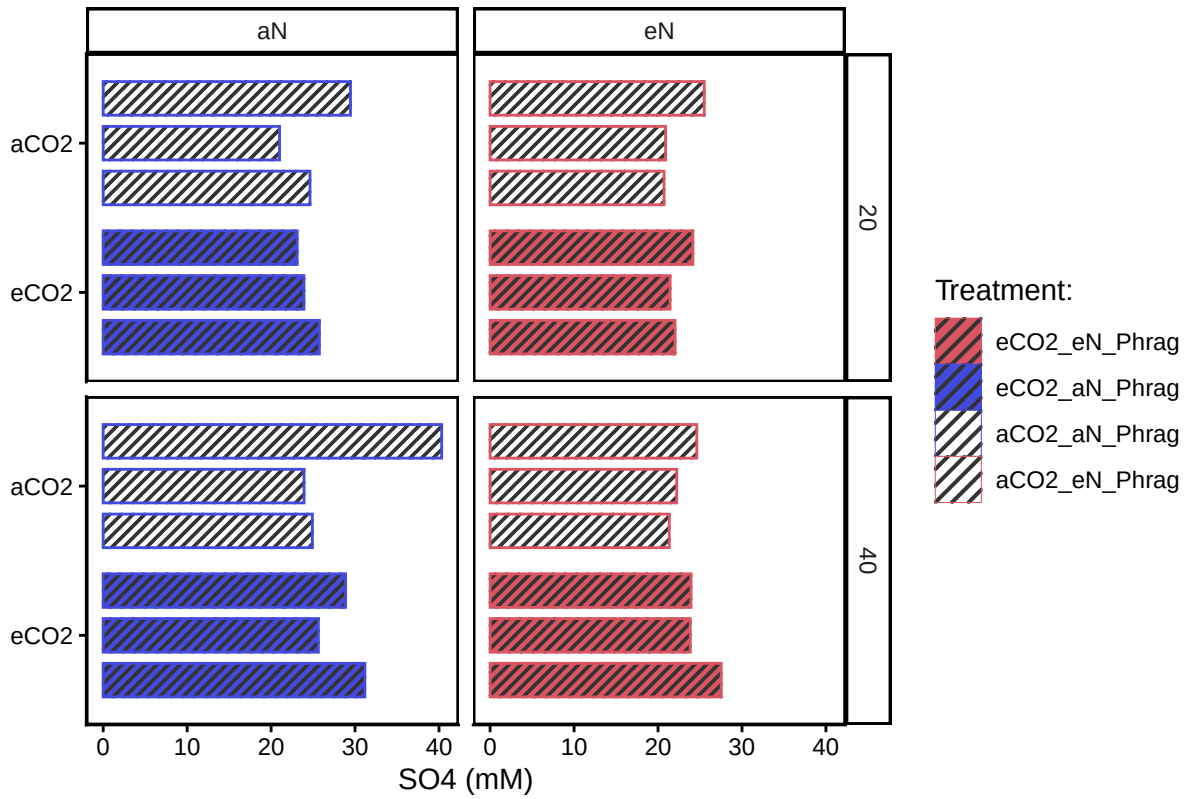


Warning: Removed 24 rows containing missing values or values outside the scale range
 ## ('geom_point()').

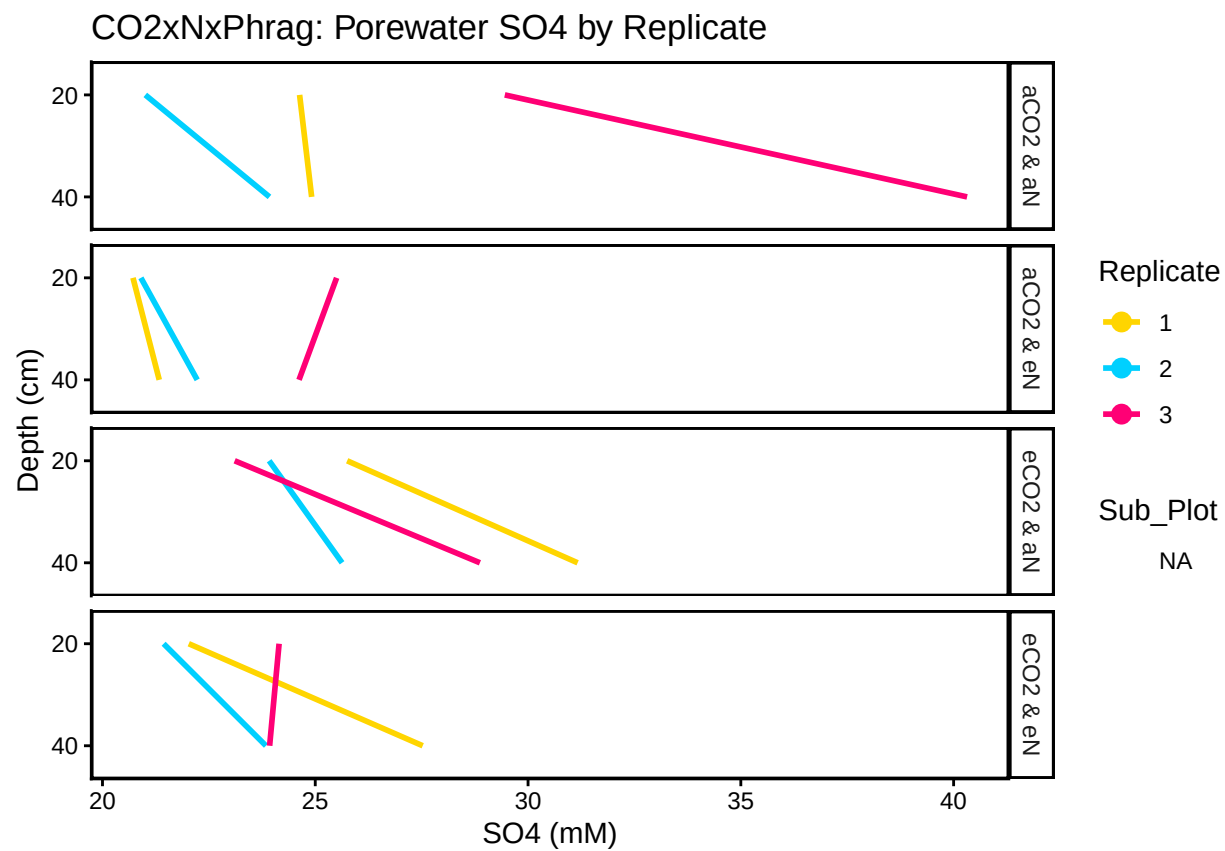


Visualize S04 Data

CO2xNxPhrag: Porewater Sulfate



Warning: Removed 24 rows containing missing values or values outside the scale range
 ## ('geom_point()').



0.11 Export Processed Data

#end